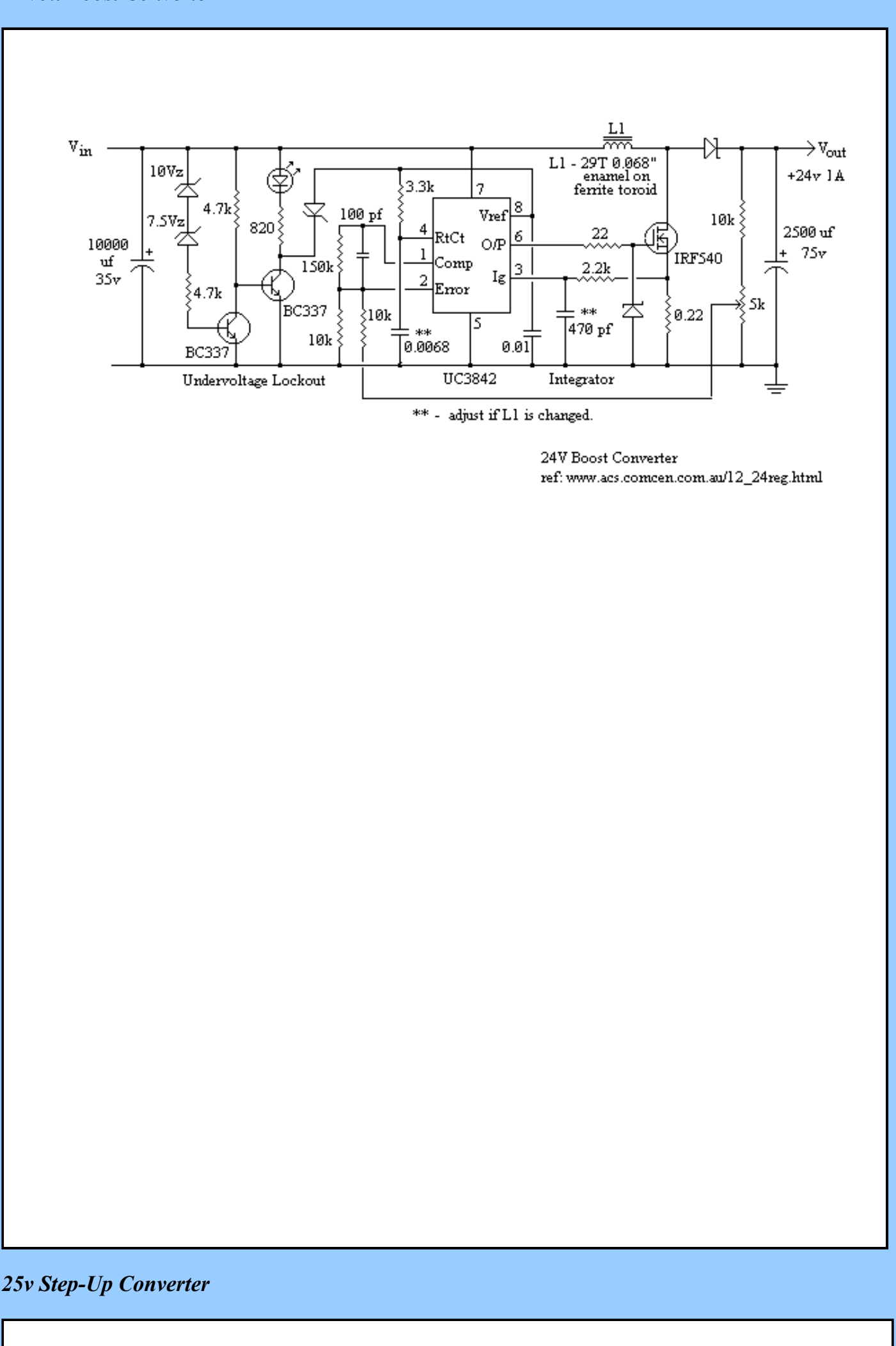
# Switch Mode and PWM Drawings

Schematics - Switching Regulators

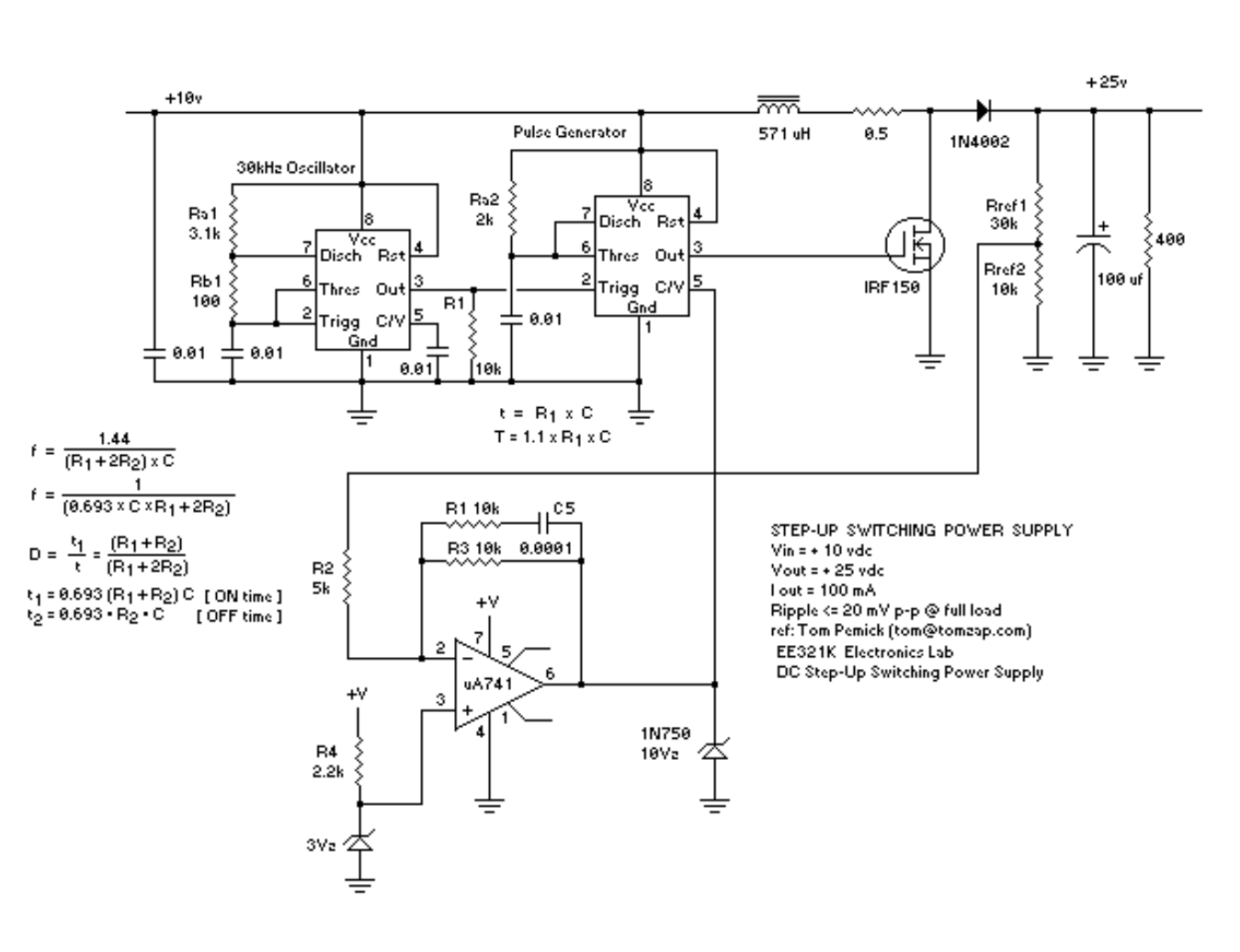
- 24 volt Boost Converter
- 25v Step-Up Converter
- Simple 2W Switching Supply
- D1415 P.S. Board
- 555-based Driver
- PWM Step-down Converter
- KA7500 (TL494) Typical Example
- NJM2355 Data Sheet Circuit
- Another NJM2355 Application
- NEC R8205A Power Supply (1 of 3)
- NEC R8205A Power Supply (2 of 3)
- NEC R8205A Power Supply (3 of 3)

***24 volt Boost Converter***


V_in
10Vz
7.5Vz
4.7k
820
4.7k
10000 uf 35v
BC337
BC337
150k
10k
100 pf
3.3k
RtCt
Comp
Error
Vref
O/P
Ig
10k
0.0068
0.01
UC3842
L1
L1 - 29T 0.068" enamel on ferrite toroid
22
2.2k
470 pf
IRF540
0.22
10k
5k
2500 uf 75v
V_out +24v 1A
Undervoltage Lockout
Integrator
** - adjust if L1 is changed.

24V Boost Converter
ref: www.acs.comcen.com.au/12_24reg.html

***25v Step-Up Converter***


+10v
30kHz Oscillator
Pulse Generator
Rs1 3.1k
Rb1 100
Rs2 2k
R1
Vcc
Dischg
Rst
Thres
Out
Trigg
C/V
Gnd
0.01
0.01
0.01
0.01
10k
t = R1 x C
T = 1.1 x R1 x C
571 uH
0.5
1N4002
+25v
Rref1 30k
Rref2 10k
100 uf
400
IRF150
R1 10k
C5
R3 10k
0.0001
R2 5k
+V
uA741
R4 2.2k
3Vz
1N750 10Vz

$f = \frac{1.44}{(R_1 + 2R_2) \times C}$

$f = \frac{1}{(0.693 \times C \times (R_1 + 2R_2))}$

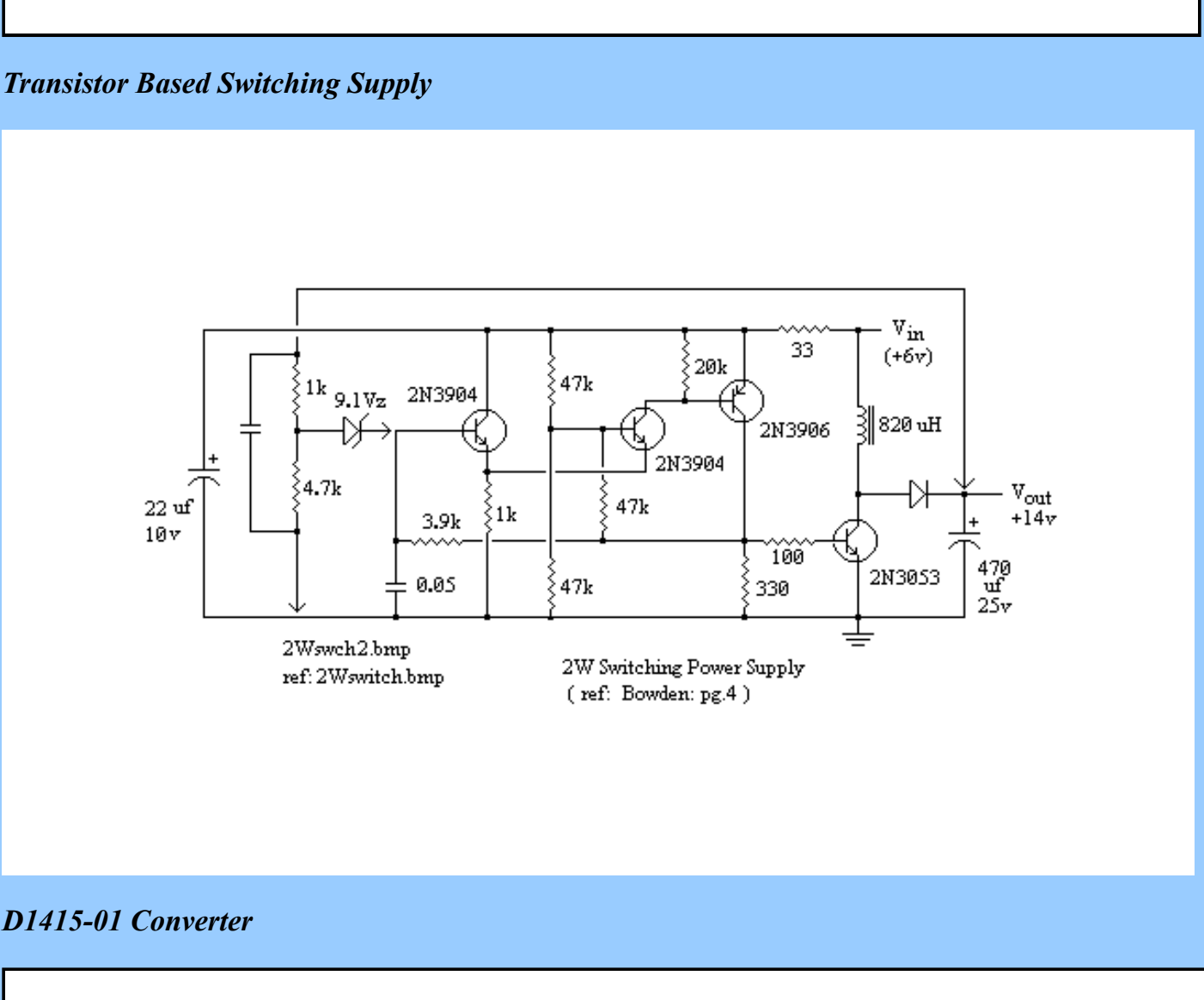
$D = \frac{t_1}{t} = \frac{(R_1 + R_2)}{(R_1 + 2R_2)}$

$t_1 = 0.693 (R_1 + R_2) C$ [ON time]

$t_2 = 0.693 \times R_2 \times C$ [OFF time]

STEP-UP SWITCHING POWER SUPPLY
Vin = + 10 vdc
Vout = + 25 vdc
I out = 100 mA
Ripple <= 20 mV p-p @ full load
ref: Tom Pumick (tom@tomzsp.com)
EE321K Electronics Lab
DC Step-Up Switching Power Supply

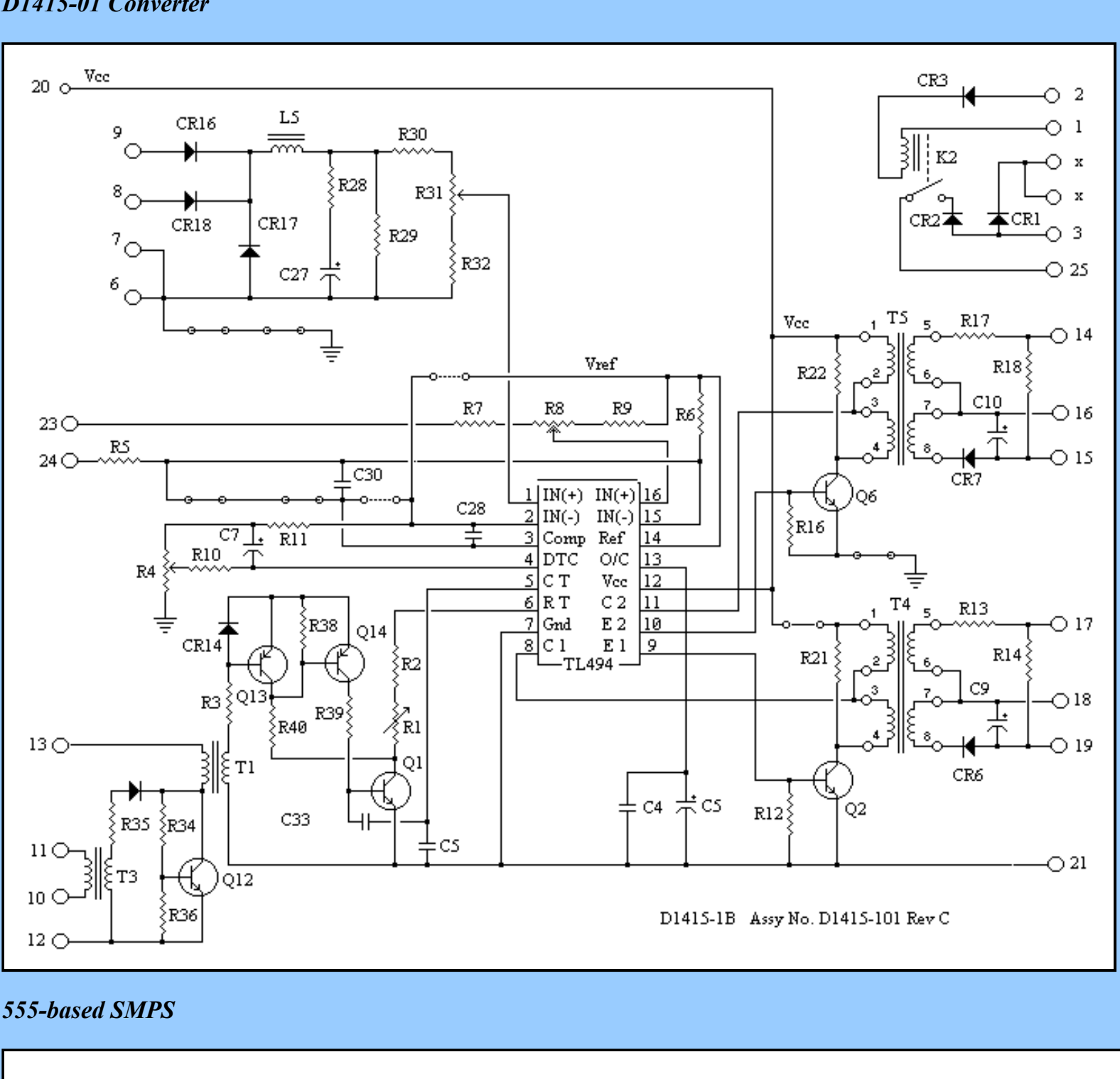
***Transistor Based Switching Supply***


1k
9.1Vz
2N3904
4.7k
22 uf 10v
3.9k
0.05
1k
47k
47k
47k
2N3904
20k
2N3906
33
V_in (+6v)
820 uH
100
330
2N3053
V_out +14v
470 uf 25v

2Wswch2.bmp
ref: 2Wswitch.bmp

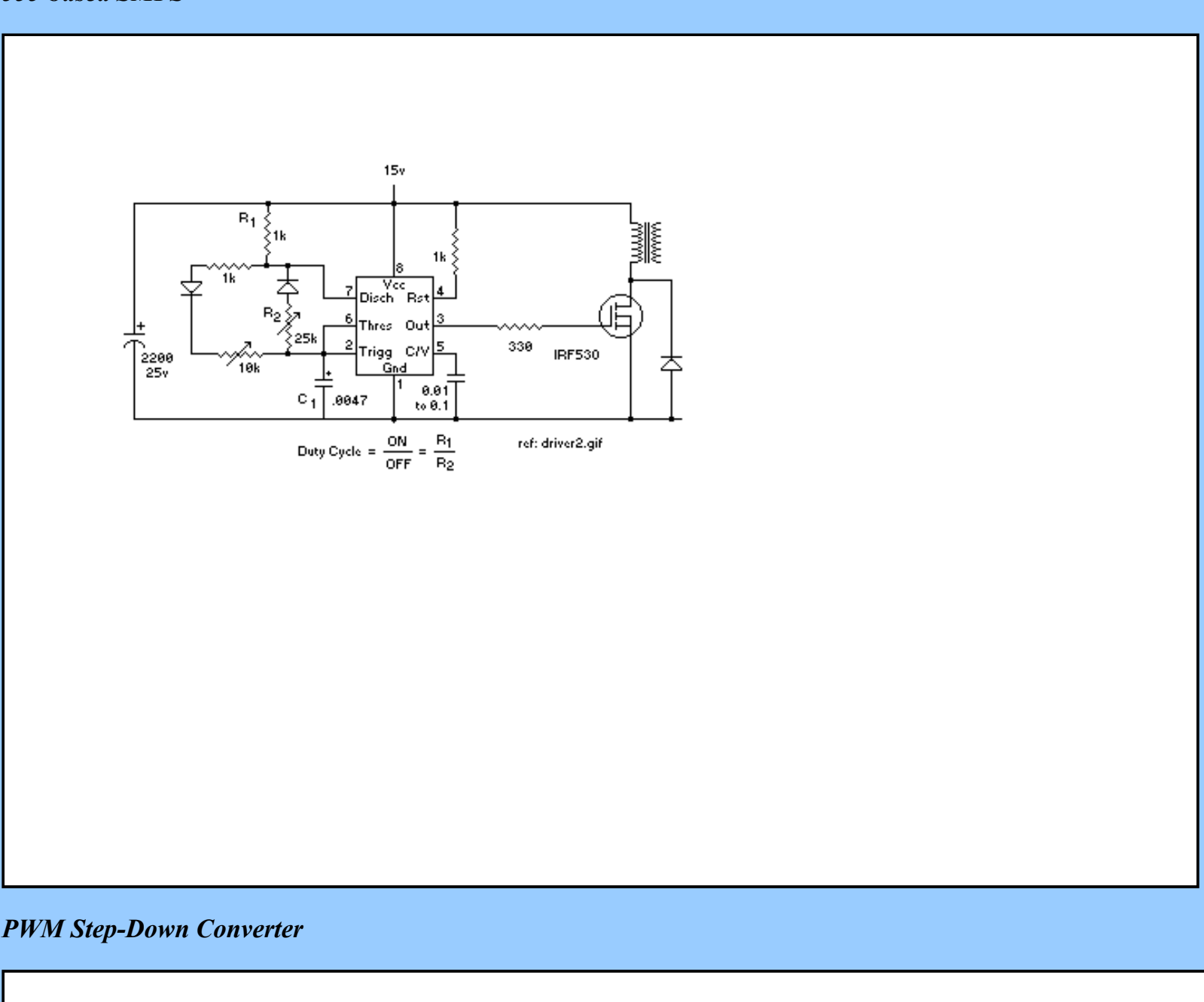
2W Switching Power Supply
( ref: Bowden: pg.4 )

***D1415-01 Converter***


20
Vcc
9
8
7
6
CR16
CR18
CR17
L5
R28
R29
R30
R31
R32
C27
CR3
K2
CR2
CR1
2
1
x
x
3
25
Vcc
T5
R17
R18
R22
C10
CR7
14
16
15
Vref
R7
R8
R9
R6
23
24
R5
C30
C28
R4
R10
C7
R11
IN(+)
IN(-)
Comp
DTC
C T
R T
Gnd
C 1
IN(+)
IN(-)
Ref
O/C
Vcc
C 2
E 2
E 1
TL494
Q6
R16
T4
R13
R14
R21
C9
CR6
17
18
19
CR14
Q13
Q14
R38
R40
R39
R3
R2
R1
Q1
13
T1
C33
C5
C4
C5
R12
Q2
21
11
10
12
R35
R34
T3
Q12
R36

D1415-1B Assy No. D1415-101 Rev C

***555-based SMPS***


15v
R1
1k
1k
R2
25k
10k
1k
Vcc
Disch
Rst
Thres
Out
Trigg
C/V
Gnd
330
IRF530
2200 25v
C1 .0047
0.01 to 0.1

$\text{Duty Cycle} = \frac{ON}{OFF} = \frac{R_1}{R_2}$

ref: drivr2.gif

***PWM Step-Down Converter***

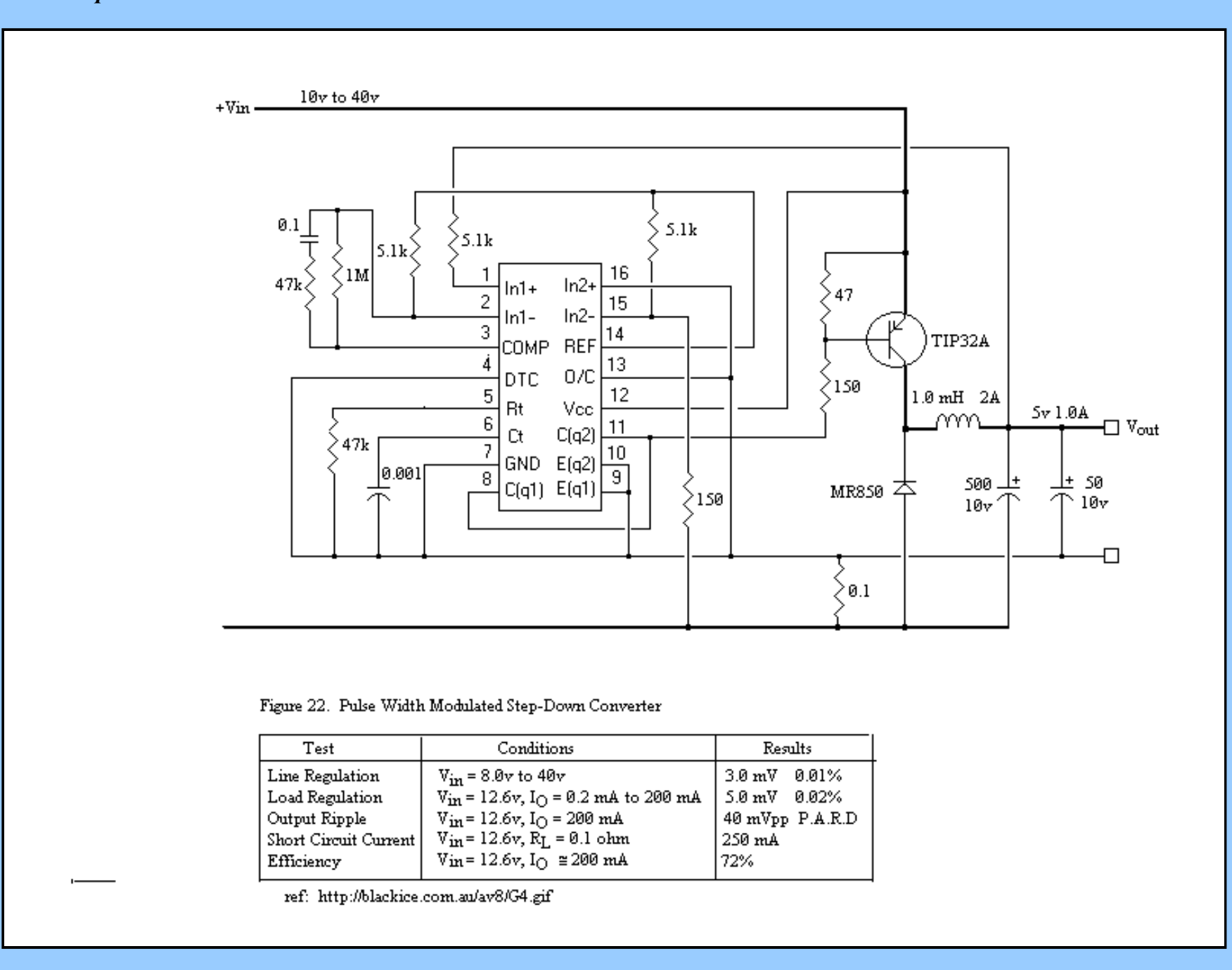
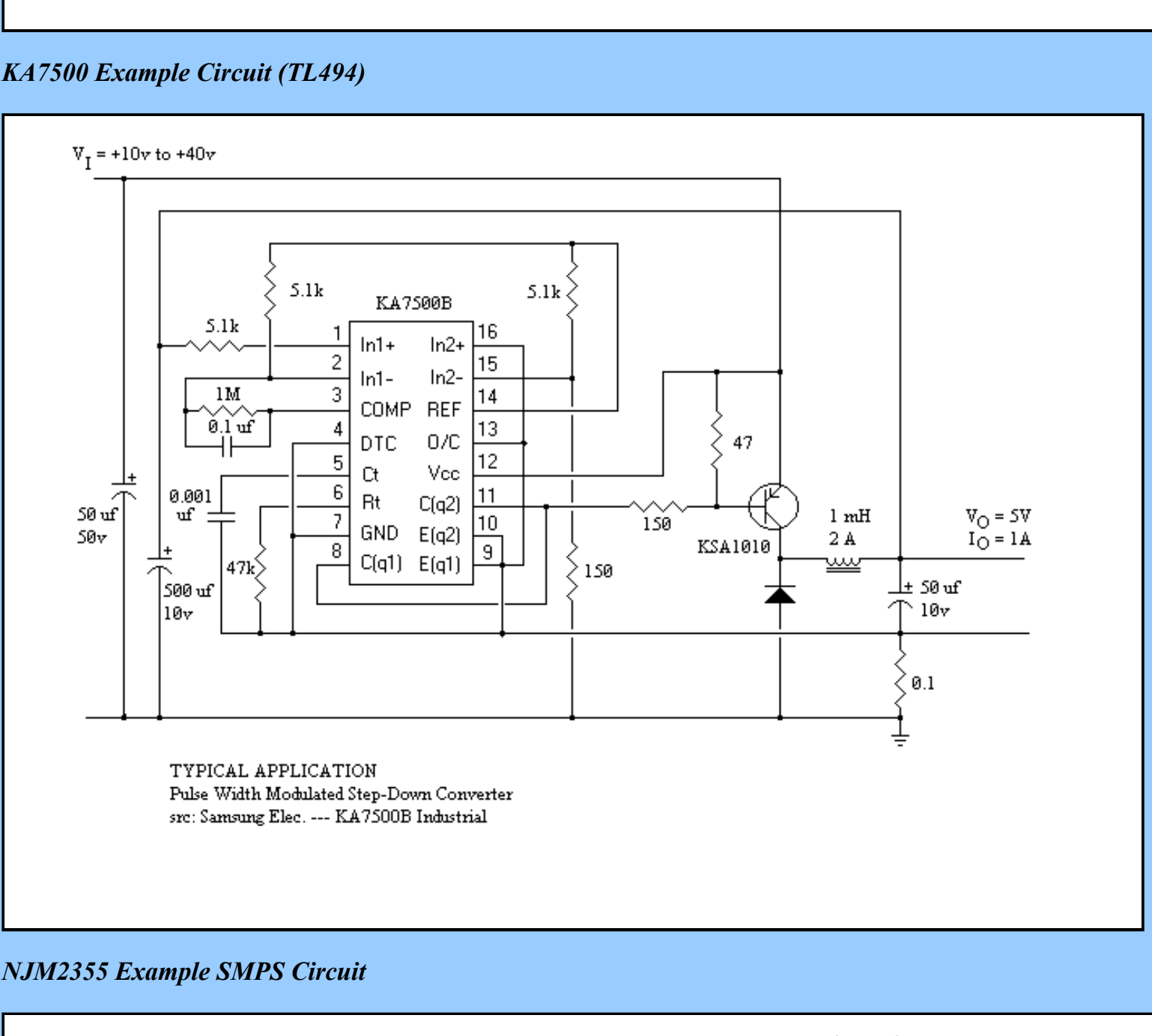

+Vin
10v to 40v
0.1
47k
1M
5.1k
5.1k
5.1k
In1+
In1-
COMP
DTC
Rt
Ct
GND
C(q1)
In2+
In2-
REF
O/C
Vcc
C(q2)
E(q2)
E(q1)
47k
0.001
47
150
TIP32A
1.0 mH
2A
5v 1.0A
Vout
MR850
500 uf 10v
50 10v
150
0.1

Figure 22. Pulse Width Modulated Step-Down Converter

| Test | Conditions | Results |
|---|---|---|
| Line Regulation | $V_{in}$ = 8.0v to 40v | 3.0 mV 0.01% |
| Load Regulation | $V_{in}$ = 12.6v, $I_O$ = 0.2 mA to 200 mA | 5.0 mV 0.02% |
| Output Ripple | $V_{in}$ = 12.6v, $I_O$ = 200 mA | 40 mVpp P.A.R.D |
| Short Circuit Current | $V_{in}$ = 12.6v, $R_L$ = 0.1 ohm | 250 mA |
| Efficiency | $V_{in}$ = 12.6v, $I_O$ = 200 mA | 72% |

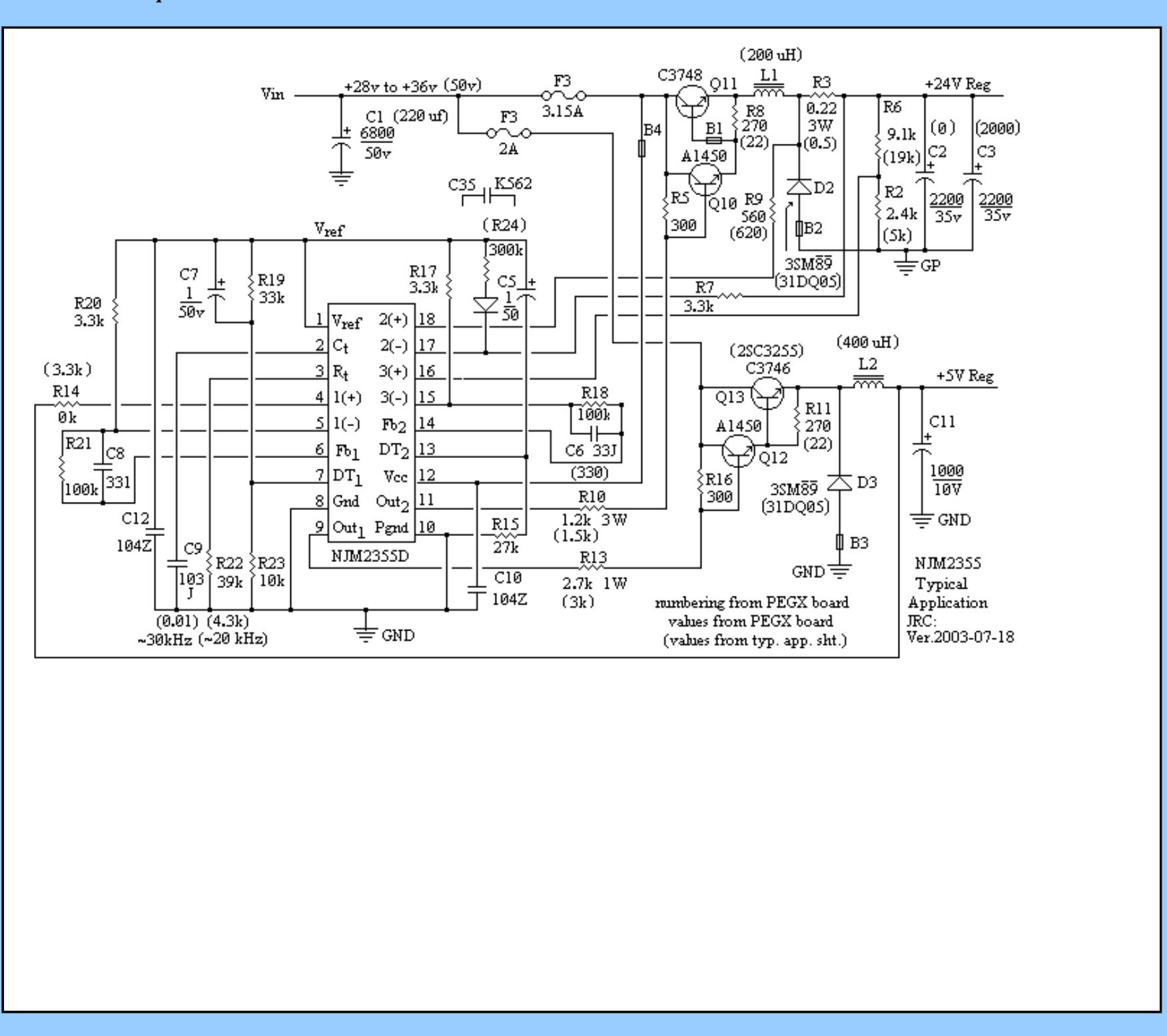
ref: http://blackice.com.au/ev8/G4.gif

***KA7500 Example Circuit (TL494)***


V_I = +10v to +40v
5.1k
5.1k
5.1k
KA7500B
In1+
In1-
COMP
DTC
Ct
Rt
GND
C(q1)
In2+
In2-
REF
O/C
Vcc
C(q2)
E(q2)
E(q1)
1M
0.1 uf
0.001 uf
47k
50 uf 50v
500 uf 10v
47
150
150
KA1010
1 mH 2 A
V_O = 5V
I_O = 1A
50 uf 10v
0.1

TYPICAL APPLICATION
Pulse Width Modulated Step-Down Converter
src: Samsung Elec. --- KA7500B Industrial

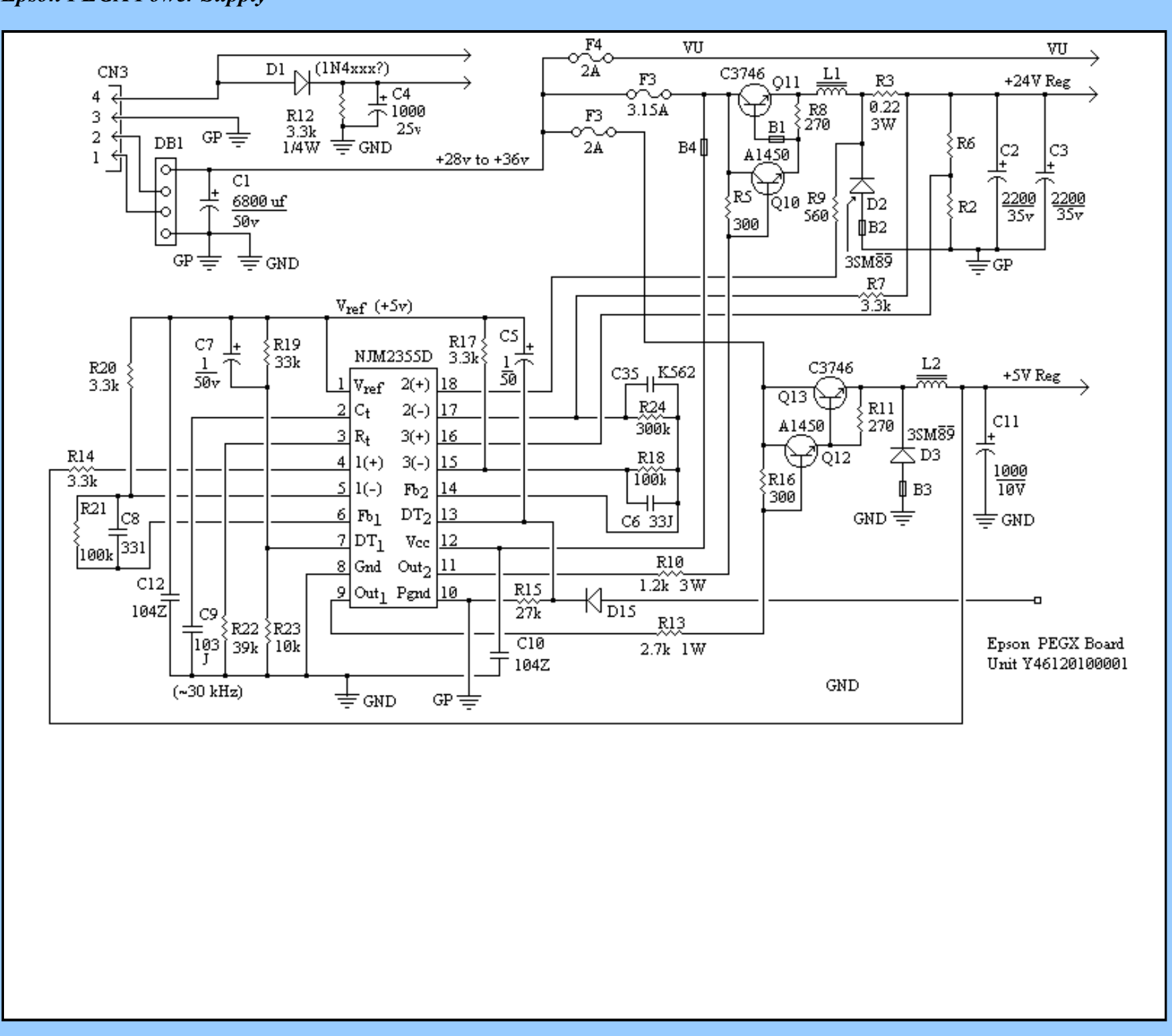
***NJM2355 Example SMPS Circuit***


Vin
+28v to +36v (50v)
C1 (220 uf) 6800 50v
F3 3.15A
F2 2A
C3748
Q11
(200 uH) L1
R3 0.22 3W (0.5)
R6 9.1k (19k)
+24V Reg
C2 (0) 2200 35v
C3 (2000) 2200 35v
R2 2.4k (5k)
GP
B4
B1
R8 270 (22)
A1450 Q10
R5 300
R9 560 (620)
D2
B2
3SM89 (31DQ05)
C35 K562
Vref
(R24) 300k
C7 1 50v
R19 33k
R17 3.3k
C5 1 50
R20 3.3k
(3.3k) R14 0k
R21 100k
C8 331
C12 104Z
C9 103
(0.01)
R22 39k (4.3k)
R23 10k
~30kHz (~20 kHz)
Vref
Ct
Rt
I(+)
I(-)
Fb1
DT1
Gnd
Out1
2(+)
2(-)
3(+)
3(-)
Fb2
DT2
Vcc
Out2
Pgnd
NJM2355D
R7 3.3k
R18 100k
C6 33J (330)
R10 1.2k 3W (1.5k)
R15 27k
C10 104Z
R13 2.7k 1W (3k)
GND
(2SC3255) C3746 Q13
A1450 Q12
R11 270 (22)
R16 300
3SM89 (31DQ05)
D3
B3
(400 uH) L2
+5V Reg
C11 1000 10V
GND

numbering from PEGX board
values from PEGX board
(values from typ. app. sht.)

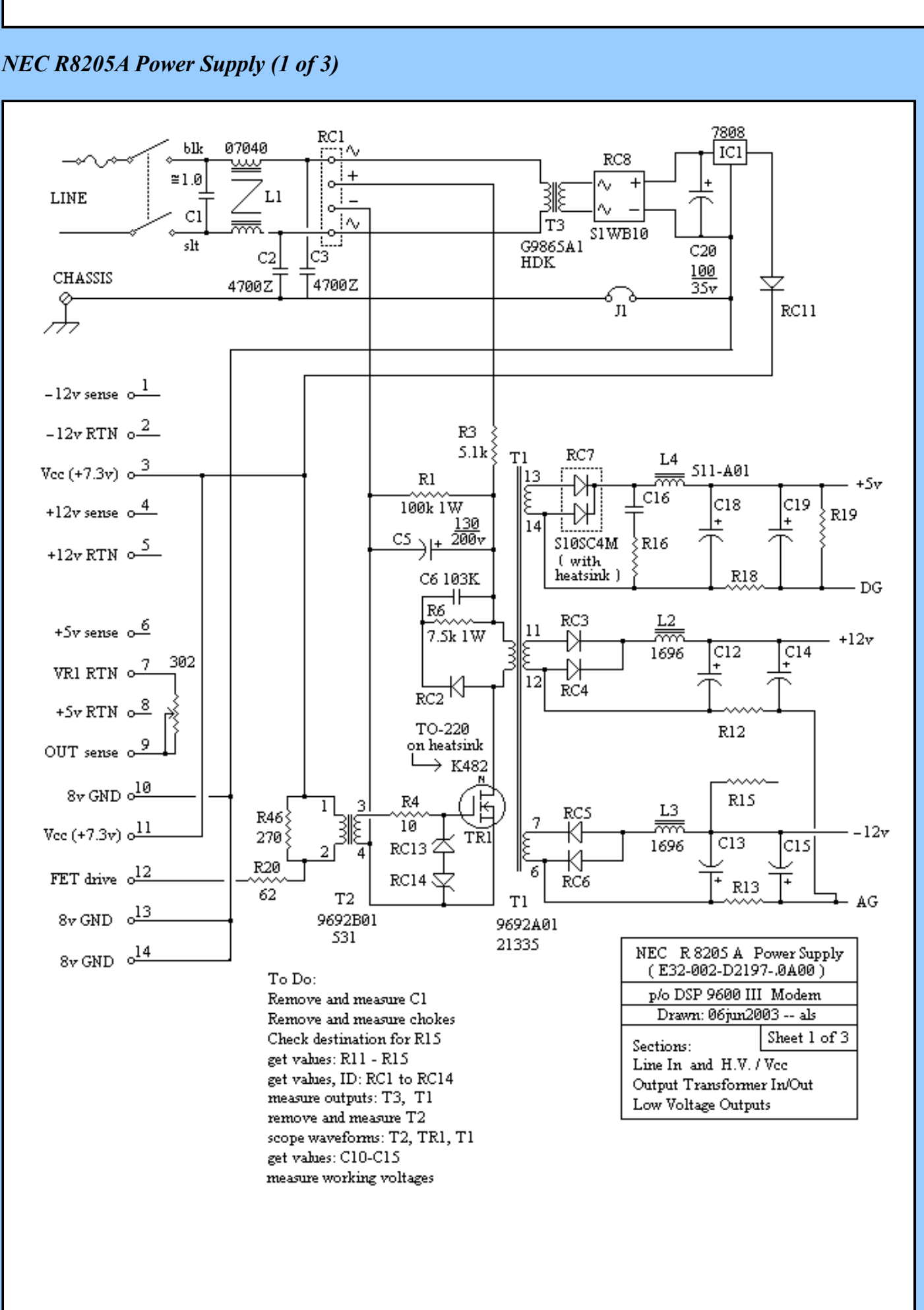
NJM2355
Typical
Application
JRC
Ver.2003-07-18

***Epson PEGX Power Supply***


CN3
D1
(1N4xxx?)
DB1
R12 3.3k 1/4W
C4 1000 25v
C1 6800 uf 50v
+28v to +36v
GP
GND
F4 2A
F3 3.15A
F3 2A
VU
C3746 Q11
L1
R3 0.22 3W
+24V Reg
B4
B1
R8 270
A1450 Q10
R5 300
R9 560
D2
B2
3SM89
R6
R2
C2 2200 35v
C3 2200 35v
R7 3.3k
Vref (+5v)
C7 1 50v
R19 33k
NJM2355D
R17 3.3k
C5 1 50
R20 3.3k
R14 3.3k
R21 100k
C8 331
C12 104Z
C9 103
R22 39k
R23 10k
(~30 kHz)
C35 K562
R24 300k
R18 100k
C6 33J
R10 1.2k 3W
R15 27k
D15
C10 104Z
R13 2.7k 1W
C3746 Q13
A1450 Q12
R11 270
3SM89
D3
B3
R16 300
L2
+5V Reg
C11 1000 10V
GND

Epson PEGX Board
Unit Y46120100001

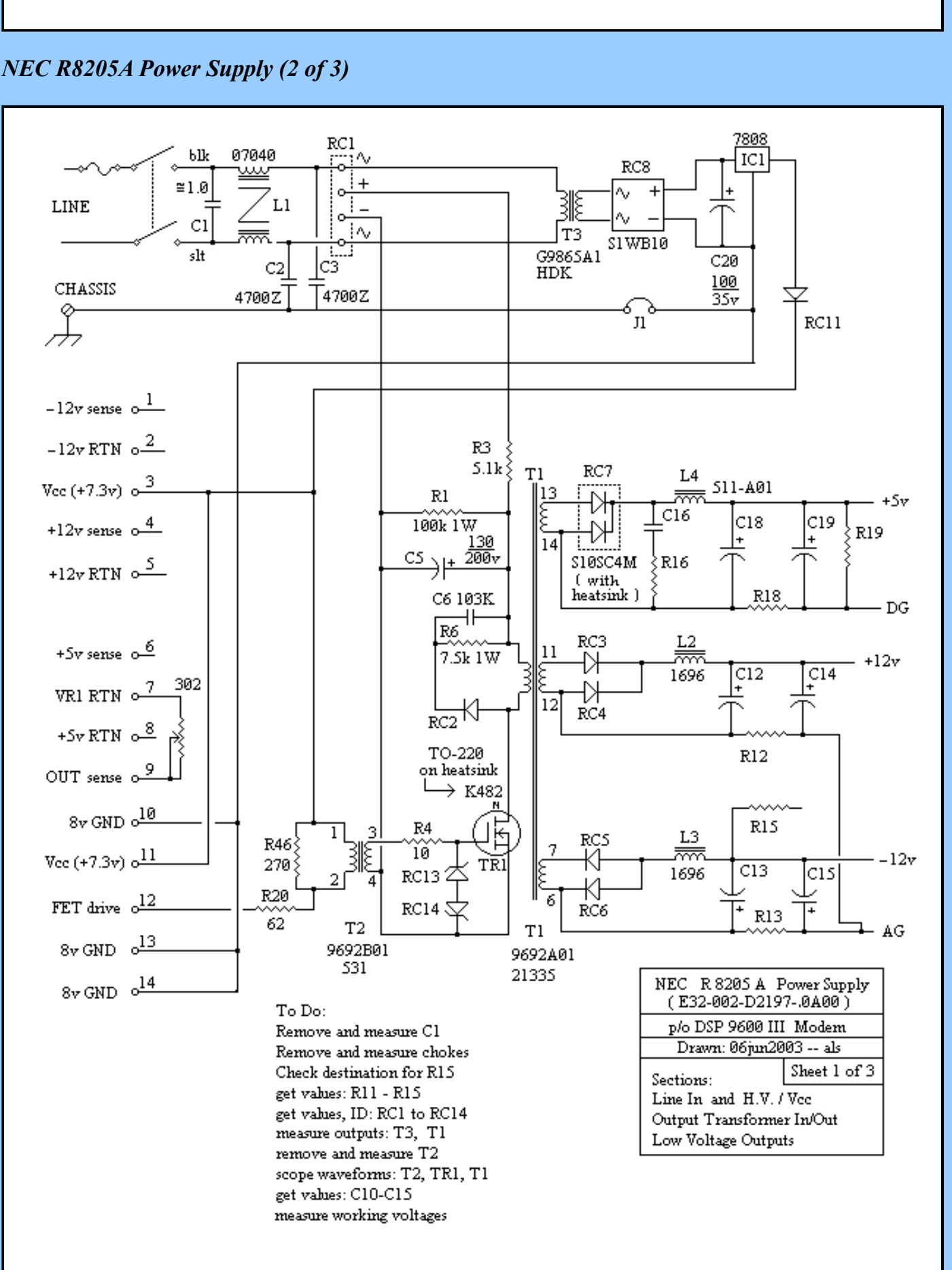
***NEC R8205A Power Supply (1 of 3)***


LINE
blk
slt
07040
#1.0
C1
L1
C2 4700Z
C3 4700Z
CHASSIS
RC1
T3 G9865A1 HDK
RC8 S1WB10
7808 IC1
C20 100 35v
J1
RC11
-12v sense 1
-12v RTN 2
Vcc (+7.3v) 3
+12v sense 4
+12v RTN 5
+5v sense 6
VR1 RTN 7
+5v RTN 8
OUT sense 9
8v GND 10
Vcc (+7.3v) 11
FET drive 12
8v GND 13
8v GND 14
302
R3 5.1k
R1 100k 1W
C5 130 200v
C6 103K
R6 7.5k 1W
RC2
TO-220 on heatsink
K482
R46 270
R20 62
T2 9692B01 531
R4 10
RC13
RC14
TR1
T1 9692A01 21335
RC7
S10SC4M ( with heatsink )
L4 S11-A01
C16
R16
C18
C19
R19
R18
+5v
DG
RC3
RC4
L2 1696
C12
C14
R12
+12v
RC5
RC6
L3 1696
R15
C13
C15
R13
-12v
AG

To Do:
Remove and measure C1
Remove and measure chokes
Check destination for R15
get values: R11 - R15
get values, ID: RC1 to RC14
measure outputs: T3, T1
remove and measure T2
scope waveforms: T2, TR1, T1
get values: C10-C15
measure working voltages

NEC R 8205 A Power Supply
( E32-002-D2197-.0A00 )
p/o DSP 9600 III Modem
Drawn: 06jun2003 -- als
Sheet 1 of 3
Sections:
Line In and H.V. / Vcc
Output Transformer In/Out
Low Voltage Outputs

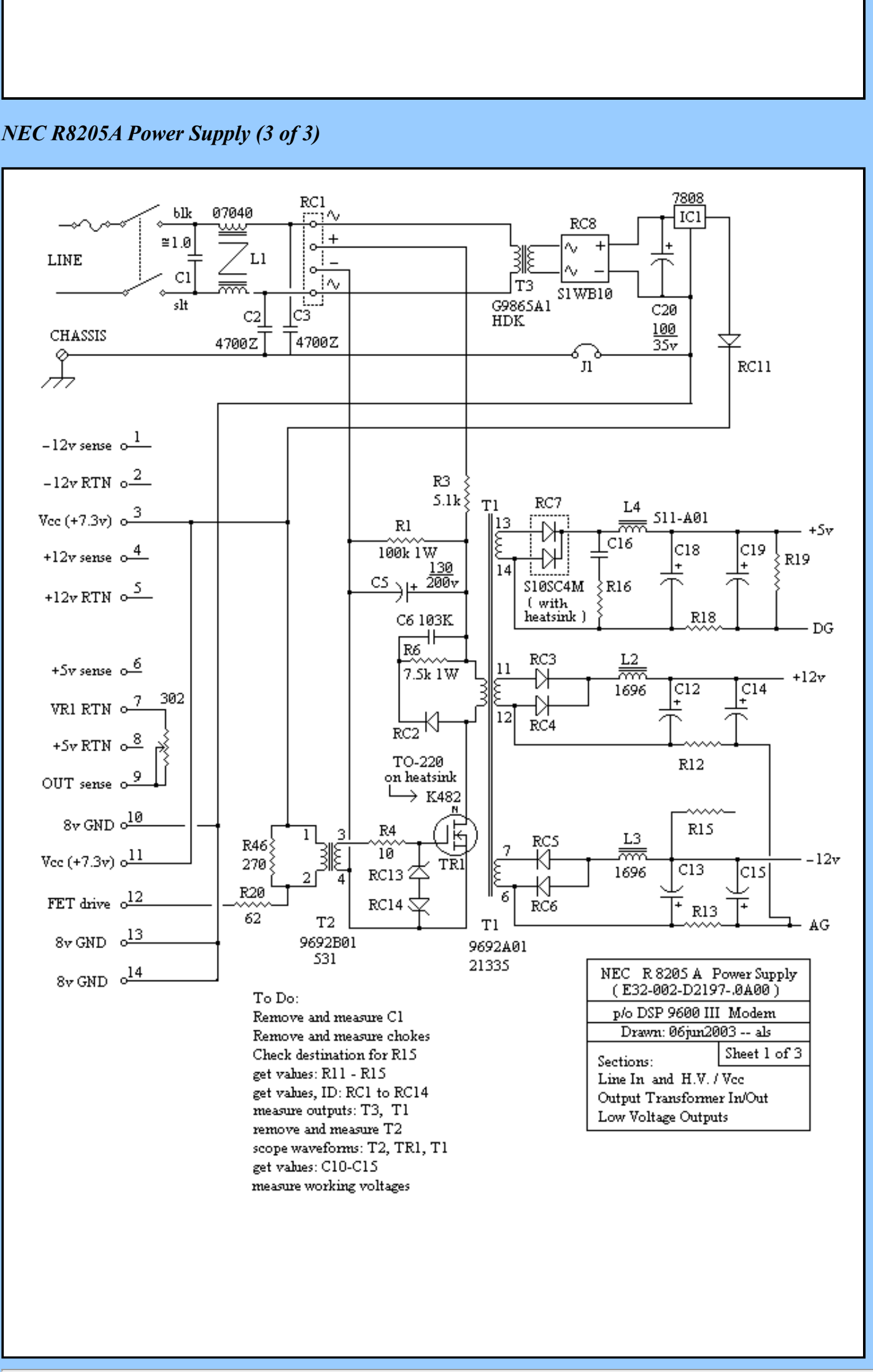
***NEC R8205A Power Supply (2 of 3)***


LINE
blk
slt
07040
#1.0
C1
L1
C2 4700Z
C3 4700Z
CHASSIS
RC1
T3 G9865A1 HDK
RC8 S1WB10
7808 IC1
C20 100 35v
J1
RC11
-12v sense 1
-12v RTN 2
Vcc (+7.3v) 3
+12v sense 4
+12v RTN 5
+5v sense 6
VR1 RTN 7
+5v RTN 8
OUT sense 9
8v GND 10
Vcc (+7.3v) 11
FET drive 12
8v GND 13
8v GND 14
302
R3 5.1k
R1 100k 1W
C5 130 200v
C6 103K
R6 7.5k 1W
RC2
TO-220 on heatsink
K482
R46 270
R20 62
T2 9692B01 531
R4 10
RC13
RC14
TR1
T1 9692A01 21335
RC7
S10SC4M ( with heatsink )
L4 S11-A01
C16
R16
C18
C19
R19
R18
+5v
DG
RC3
RC4
L2 1696
C12
C14
R12
+12v
RC5
RC6
L3 1696
R15
C13
C15
R13
-12v
AG

To Do:
Remove and measure C1
Remove and measure chokes
Check destination for R15
get values: R11 - R15
get values, ID: RC1 to RC14
measure outputs: T3, T1
remove and measure T2
scope waveforms: T2, TR1, T1
get values: C10-C15
measure working voltages

NEC R 8205 A Power Supply
( E32-002-D2197-.0A00 )
p/o DSP 9600 III Modem
Drawn: 06jun2003 -- als
Sheet 1 of 3
Sections:
Line In and H.V. / Vcc
Output Transformer In/Out
Low Voltage Outputs

***NEC R8205A Power Supply (3 of 3)***


LINE
blk
slt
07040
#1.0
C1
L1
C2 4700Z
C3 4700Z
CHASSIS
RC1
T3 G9865A1 HDK
RC8 S1WB10
7808 IC1
C20 100 35v
J1
RC11
-12v sense 1
-12v RTN 2
Vcc (+7.3v) 3
+12v sense 4
+12v RTN 5
+5v sense 6
VR1 RTN 7
+5v RTN 8
OUT sense 9
8v GND 10
Vcc (+7.3v) 11
FET drive 12
8v GND 13
8v GND 14
302
R3 5.1k
R1 100k 1W
C5 130 200v
C6 103K
R6 7.5k 1W
RC2
TO-220 on heatsink
K482
R46 270
R20 62
T2 9692B01 531
R4 10
RC13
RC14
TR1
T1 9692A01 21335
RC7
S10SC4M ( with heatsink )
L4 S11-A01
C16
R16
C18
C19
R19
R18
+5v
DG
RC3
RC4
L2 1696
C12
C14
R12
+12v
RC5
RC6
L3 1696
R15
C13
C15
R13
-12v
AG

To Do:
Remove and measure C1
Remove and measure chokes
Check destination for R15
get values: R11 - R15
get values, ID: RC1 to RC14
measure outputs: T3, T1
remove and measure T2
scope waveforms: T2, TR1, T1
get values: C10-C15
measure working voltages

NEC R 8205 A Power Supply
( E32-002-D2197-.0A00 )
p/o DSP 9600 III Modem
Drawn: 06jun2003 -- als
Sheet 1 of 3
Sections:
Line In and H.V. / Vcc
Output Transformer In/Out
Low Voltage Outputs

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